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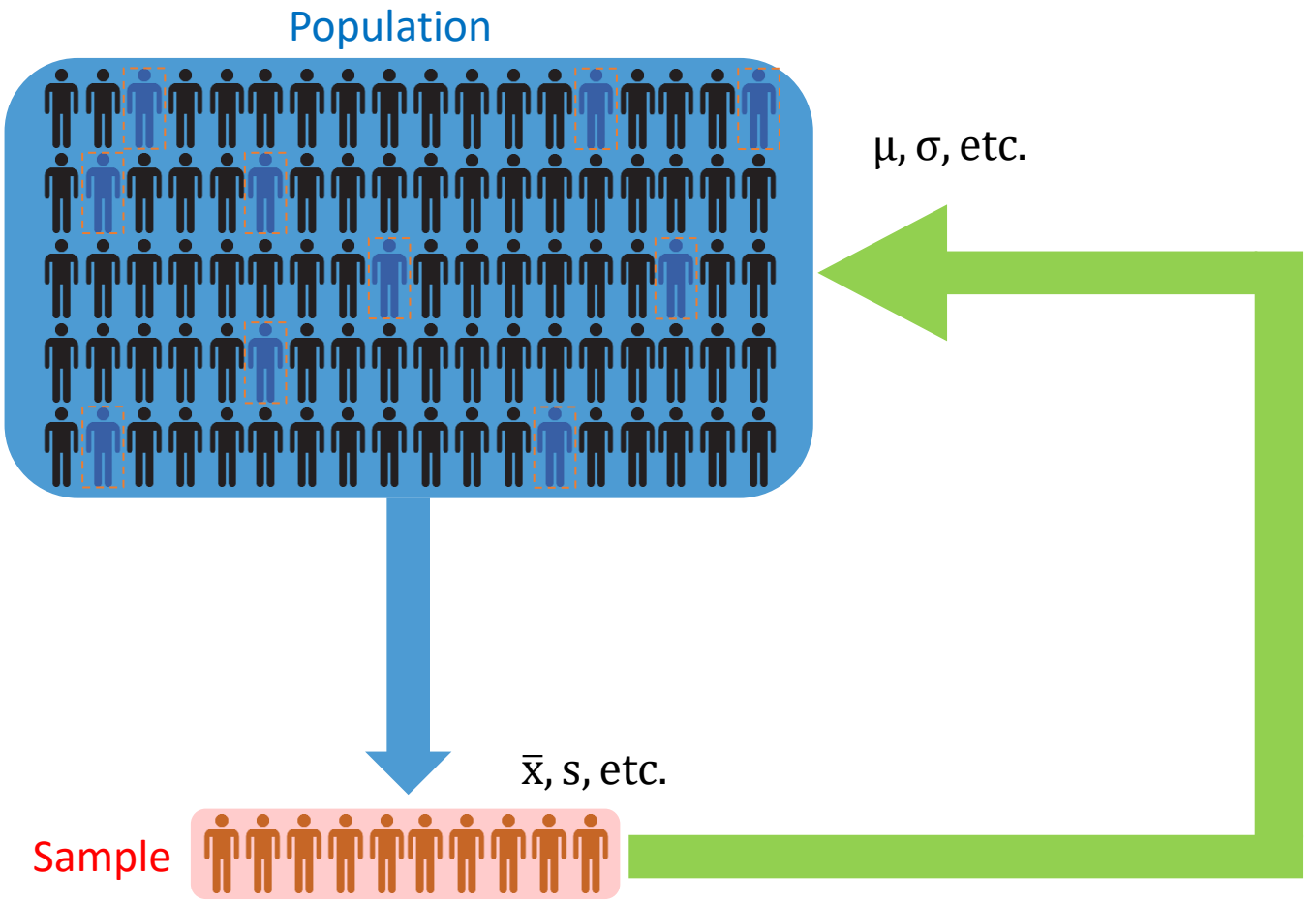
STATISTICS

Session 5: Estimation

Populations and Samples

Estimation is a fundamental part of statistics:

- we use what we can observe (the sample)
- to learn about what we cannot observe directly (the population).



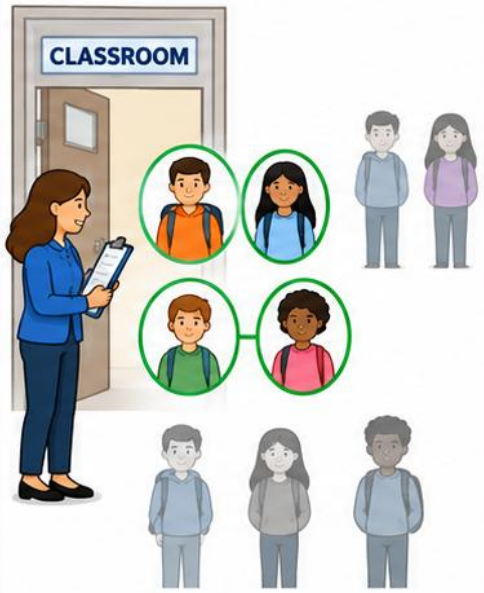
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A good sampling method can reduce error in our estimates

Observed value = True value + Systematic Error + Random Error

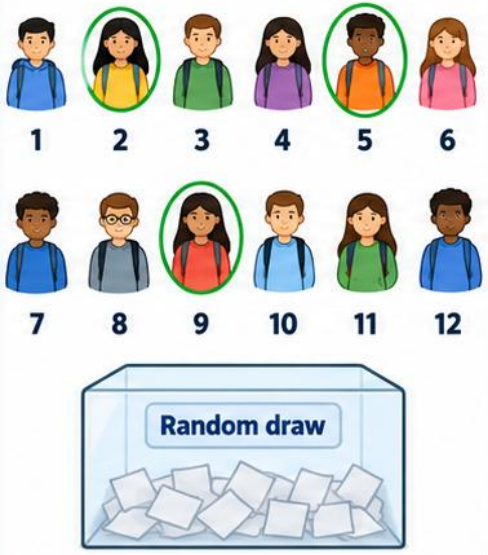
1 Convenience sampling

Select people who are easy to reach.



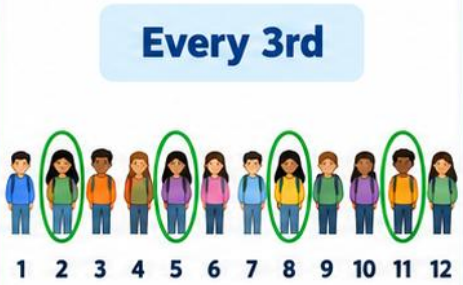
2 Simple random sampling (SRS)

Every person has an equal chance of selection.



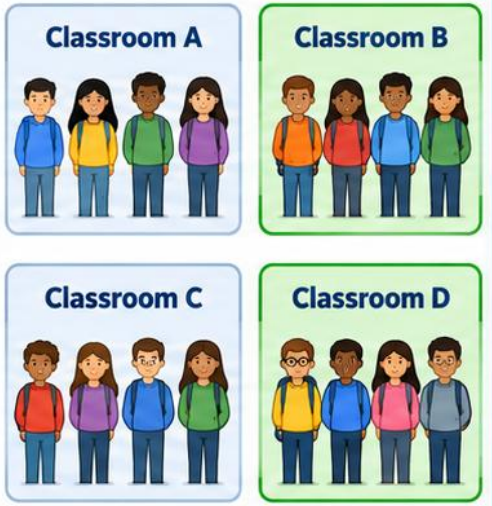
3 Systematic sampling

After a random start, select every k^{th} person.



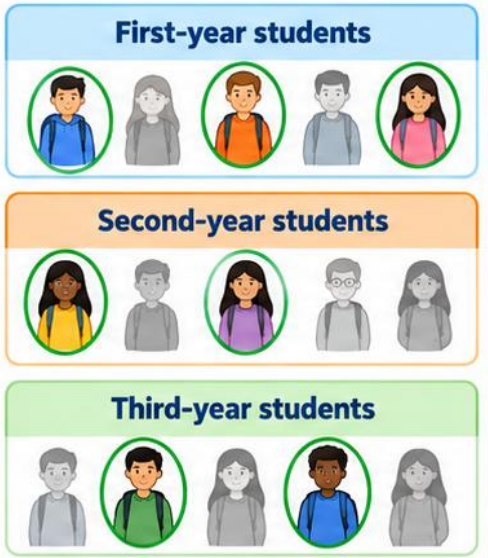
4 Cluster sampling

Randomly select groups, then include everyone in those groups.



5 Stratified sampling

Divide into groups, then randomly select people from each group.

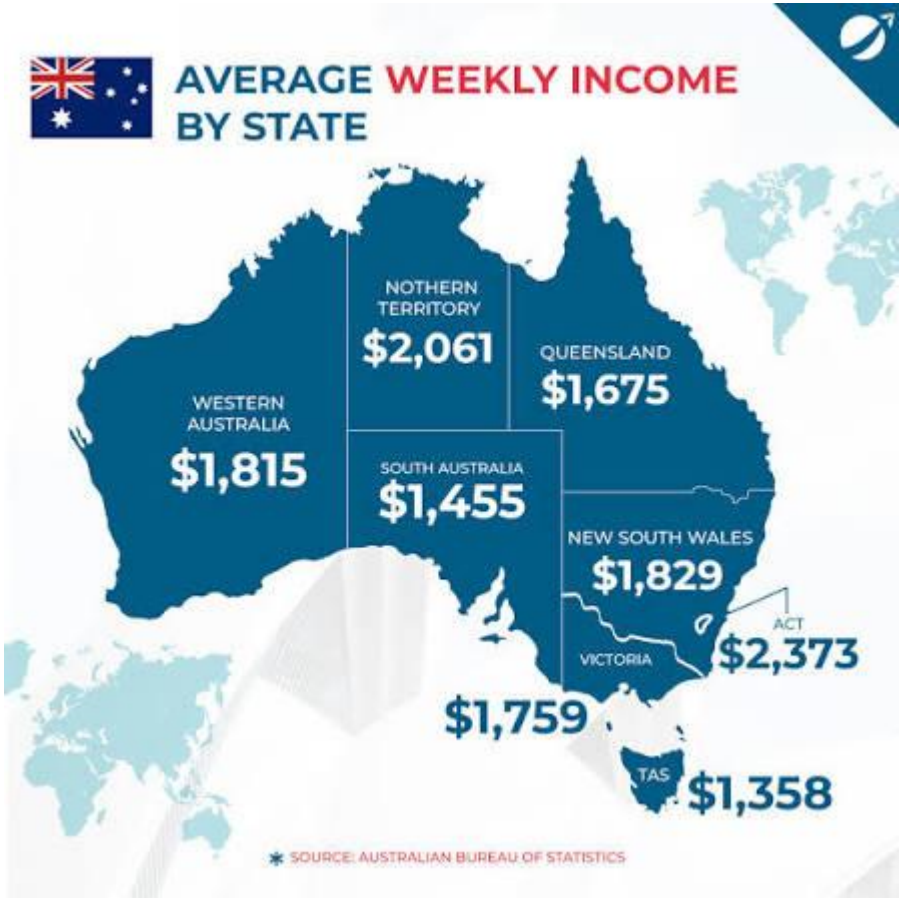


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Populations and Samples

Suppose you work for the Australian Bureau of Statistics. You have been asked to report the typical annual income of Australian households. This sounds simple, but it immediately raises statistical questions:

- What counts as a household?
- What counts as income?
- Are we interested in the mean, the median, or the proportion of households below a particular income level?
- If we only survey a sample of households, how close will our answer be to the truth?

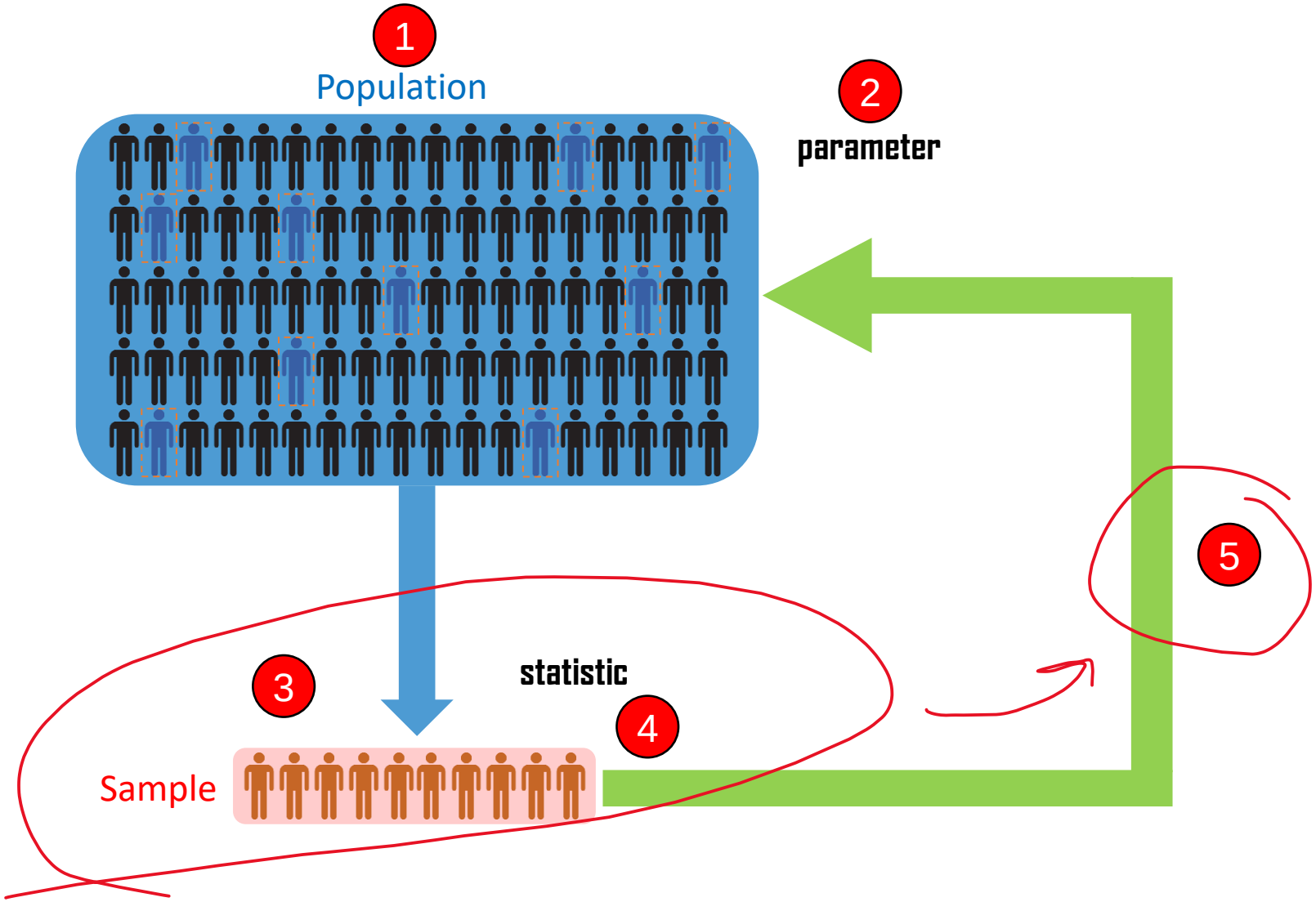


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But which statistic should we use?

The mean isn't automatically the right statistic. It depends on the question we're asking and the characteristic of the population that we're trying to estimate.”

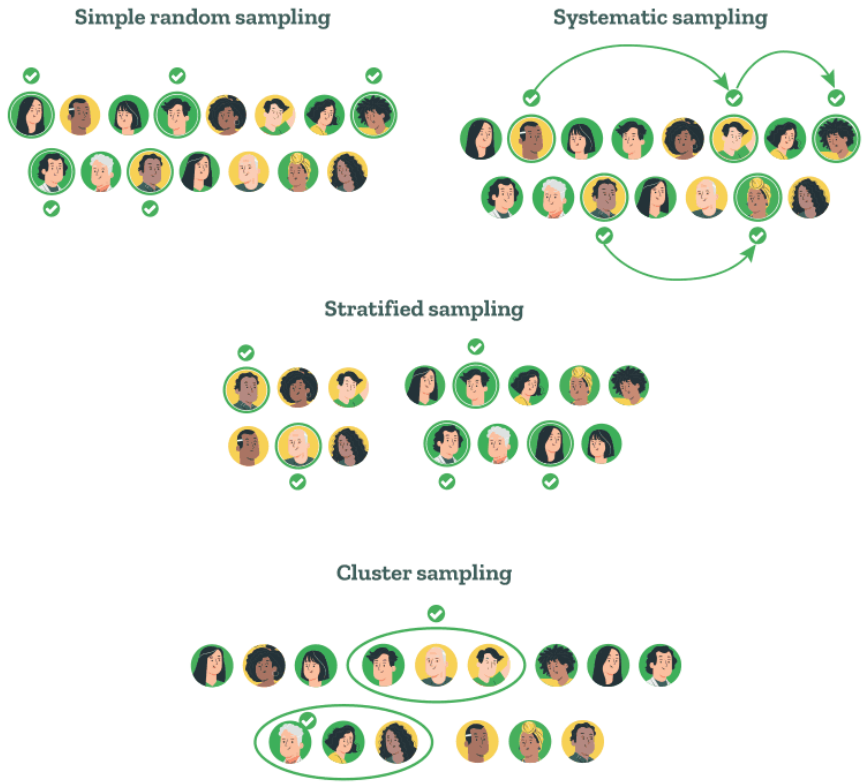
Parameters and Statistics

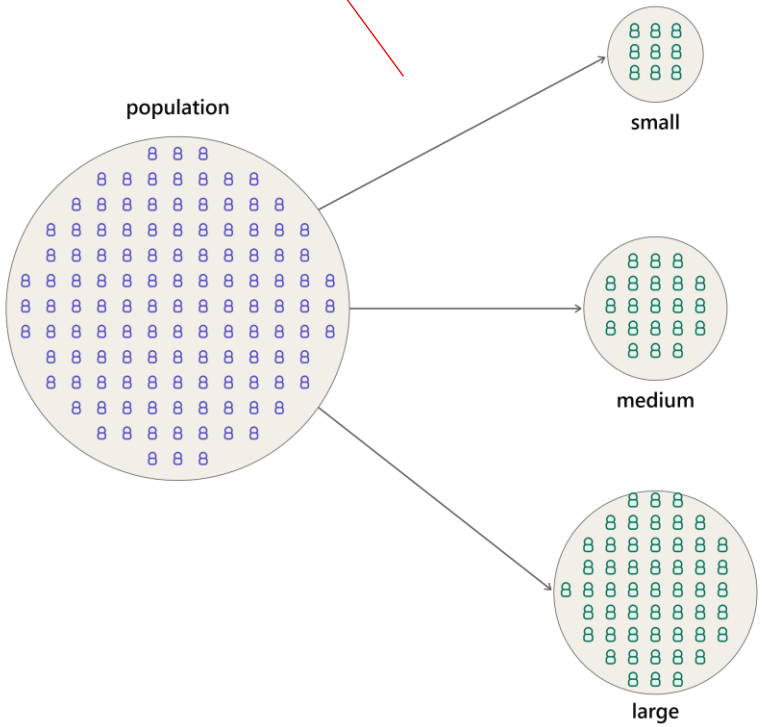


If we had chosen a different sample, would we have the same estimate?

Parameters and Statistics

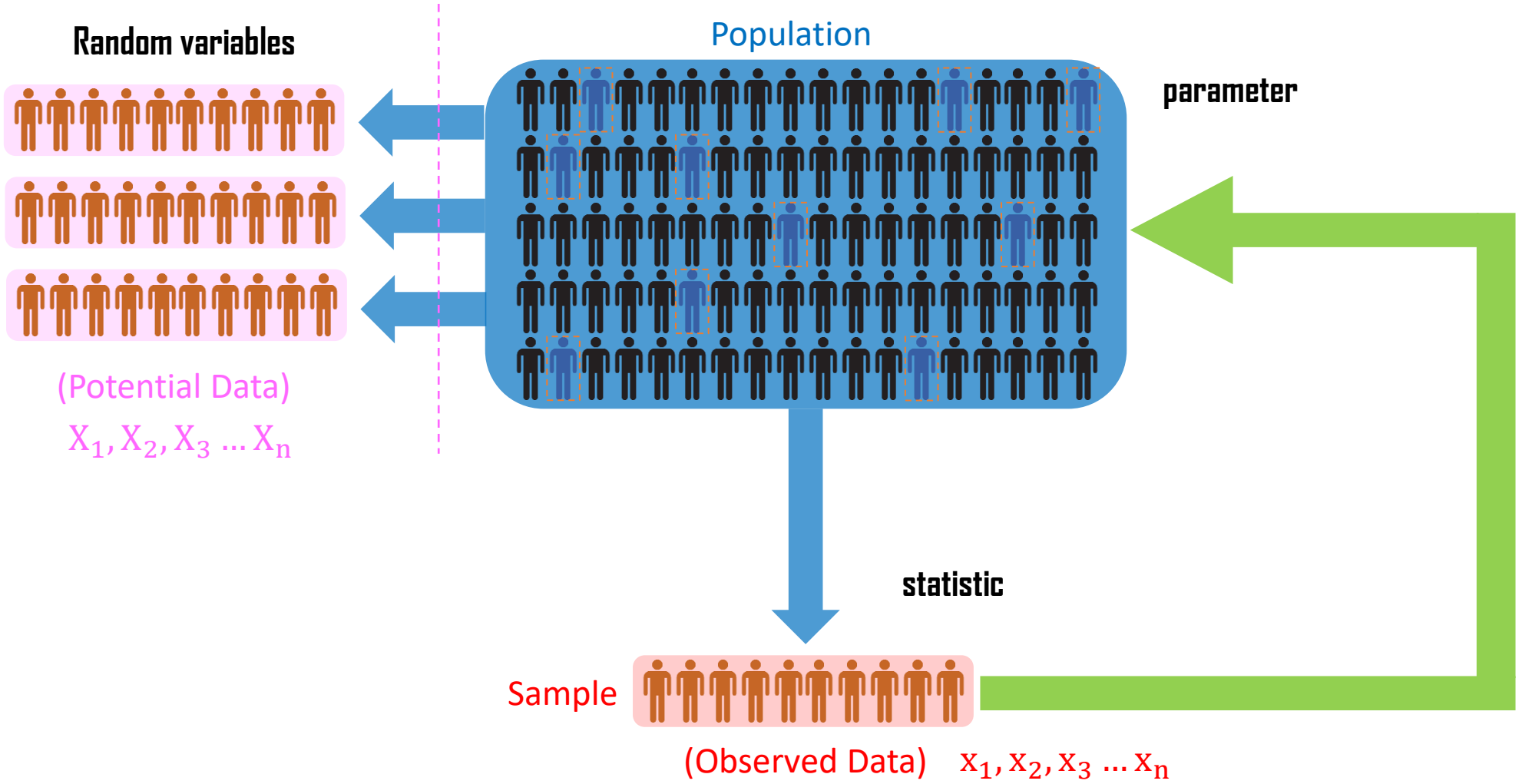
Observed value = True value + Systematic Error + Random Error



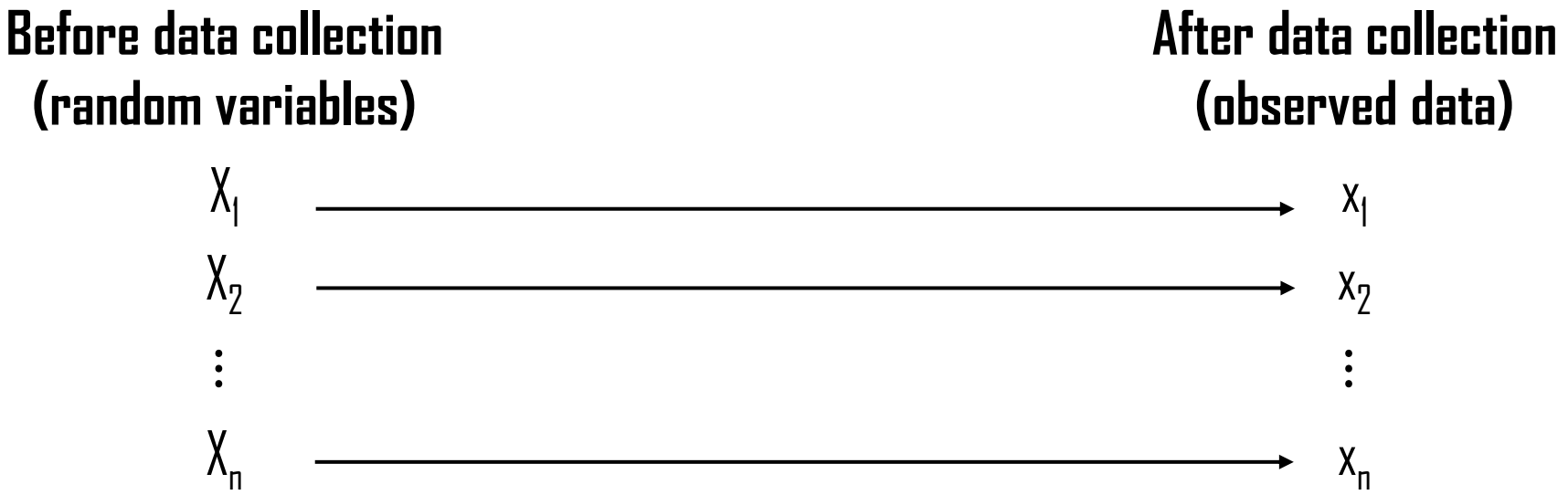


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Random variables



Random variables



Probability Distributions

X

RANDOM VARIABLE

PROBABILITY DISTRIBUTION

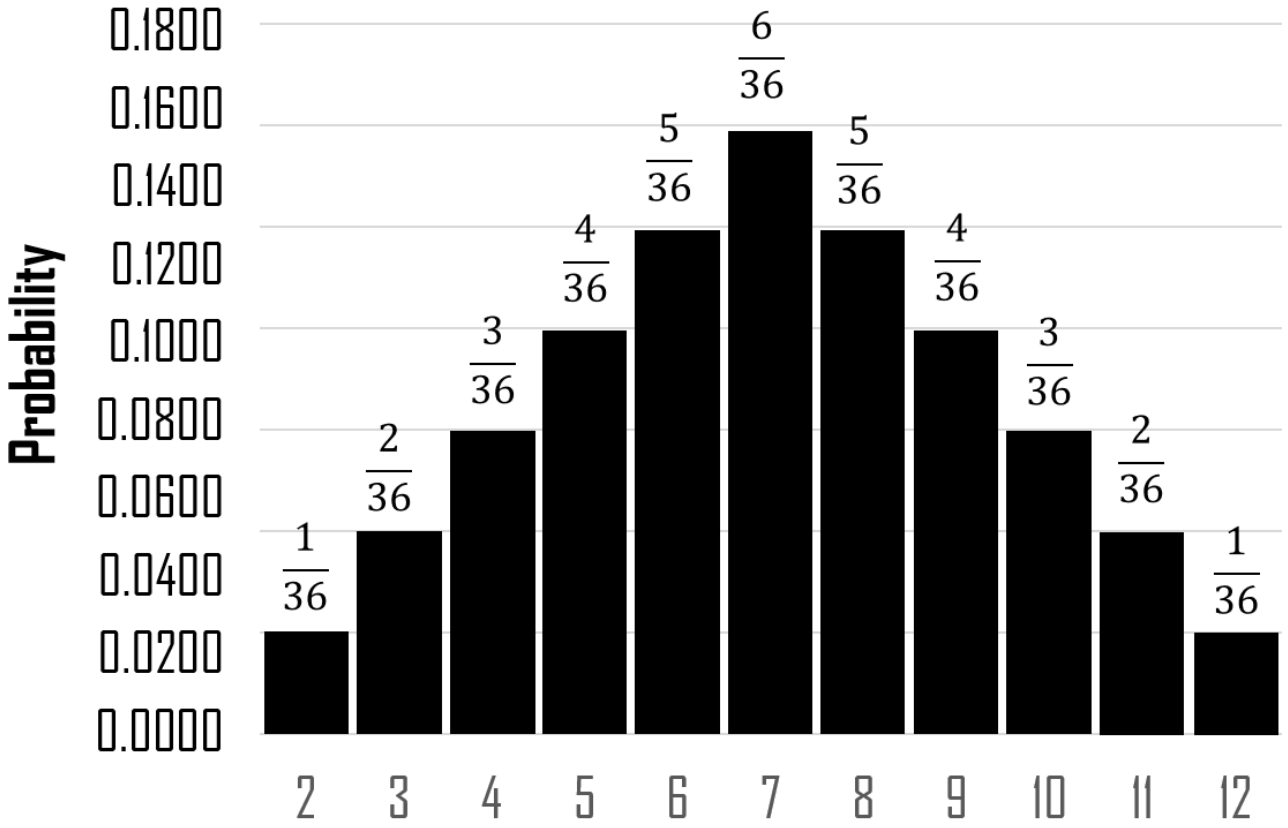
$f(x)$

PROBABILITY DENSITY FUNCTION

PROBABILITY MASS FUNCTION

Probability Distributions - Discrete

						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	12



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Probability Distributions - Continuous



$73,251

$87,000

$90,500

$91,000

$93,256.50

$93,256.6365

$93,256.637

$93,257

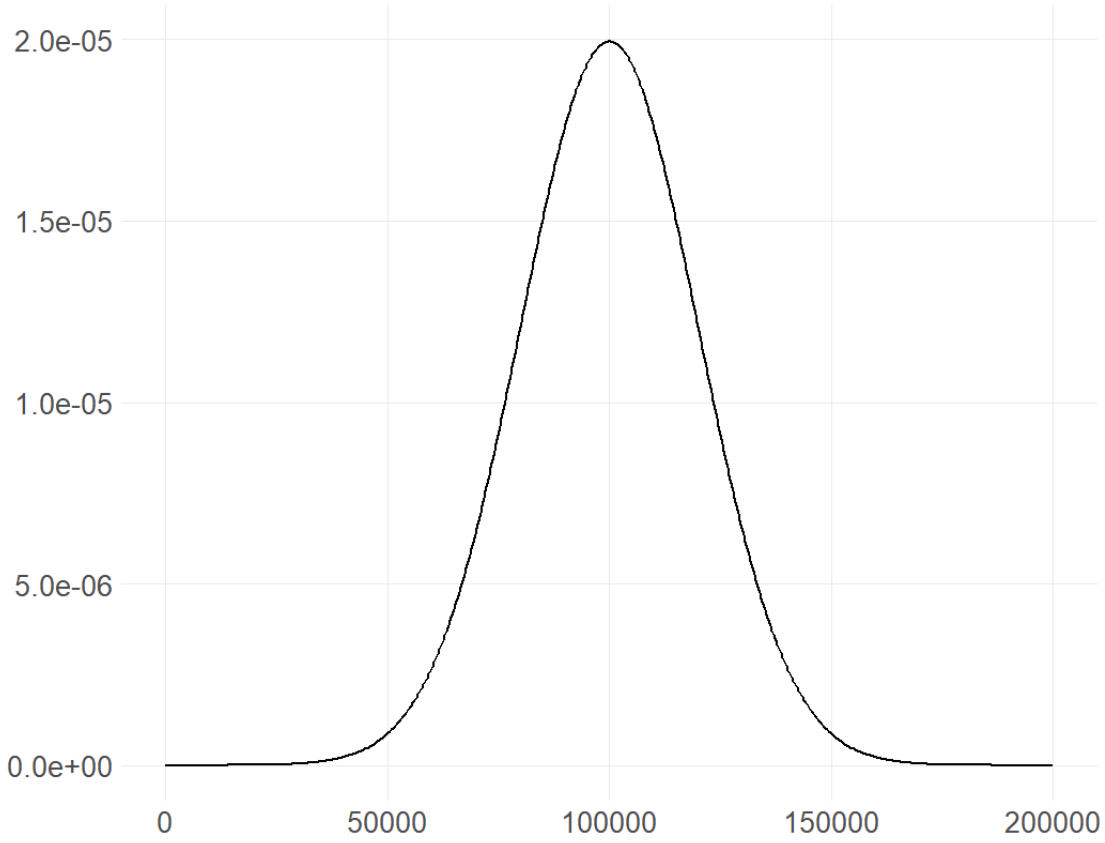
$95,000

$100,000

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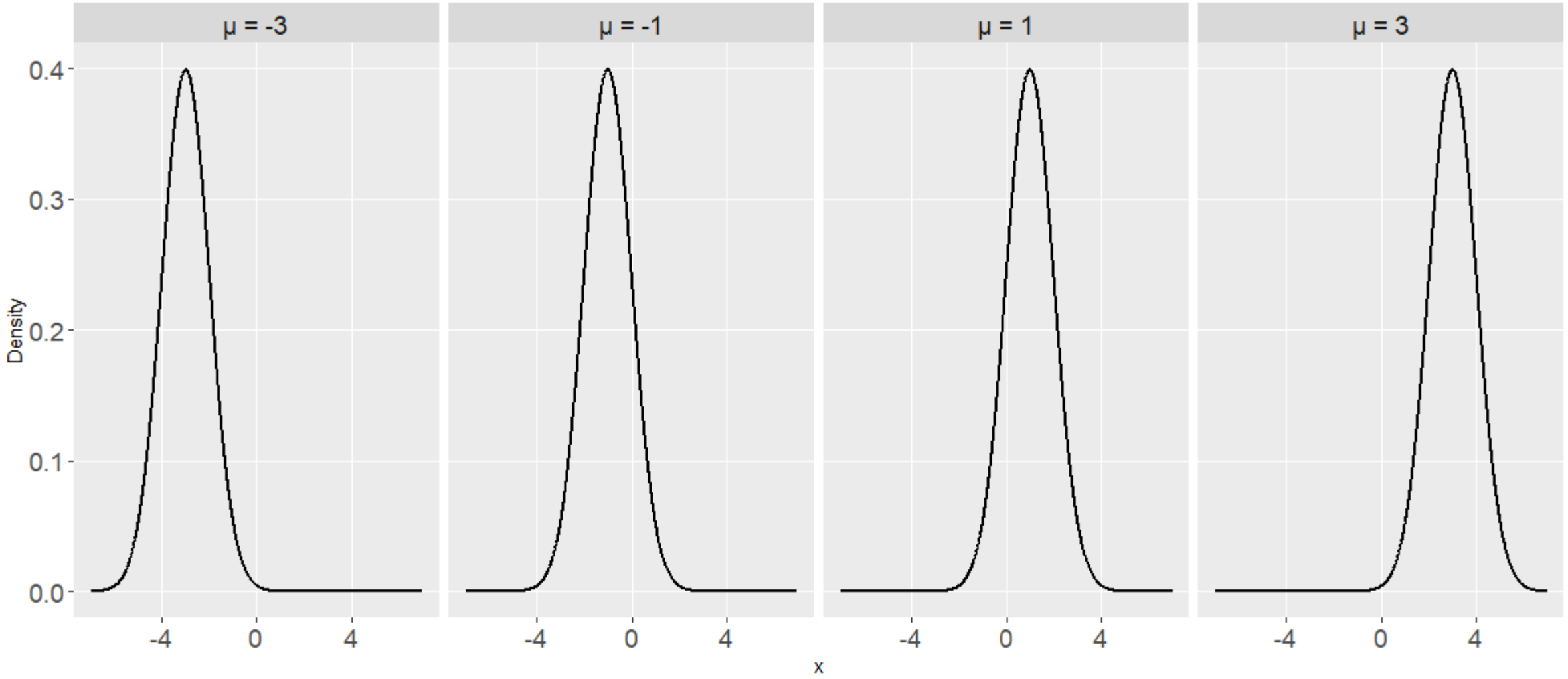
Probability Distributions - Continuous





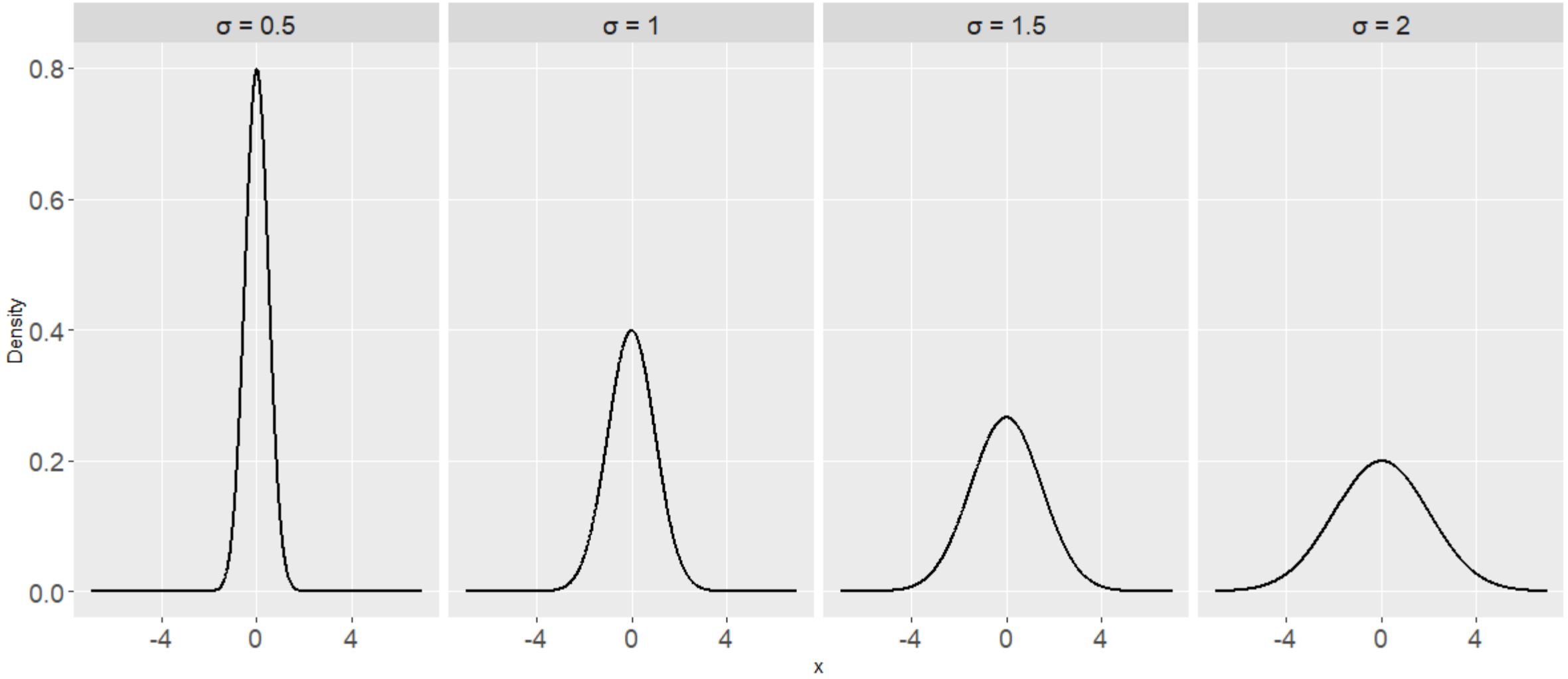
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Normal Distribution – μ



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Normal Distribution – σ



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Sampling Distribution

Our one sample gives us one observed value of our estimate:

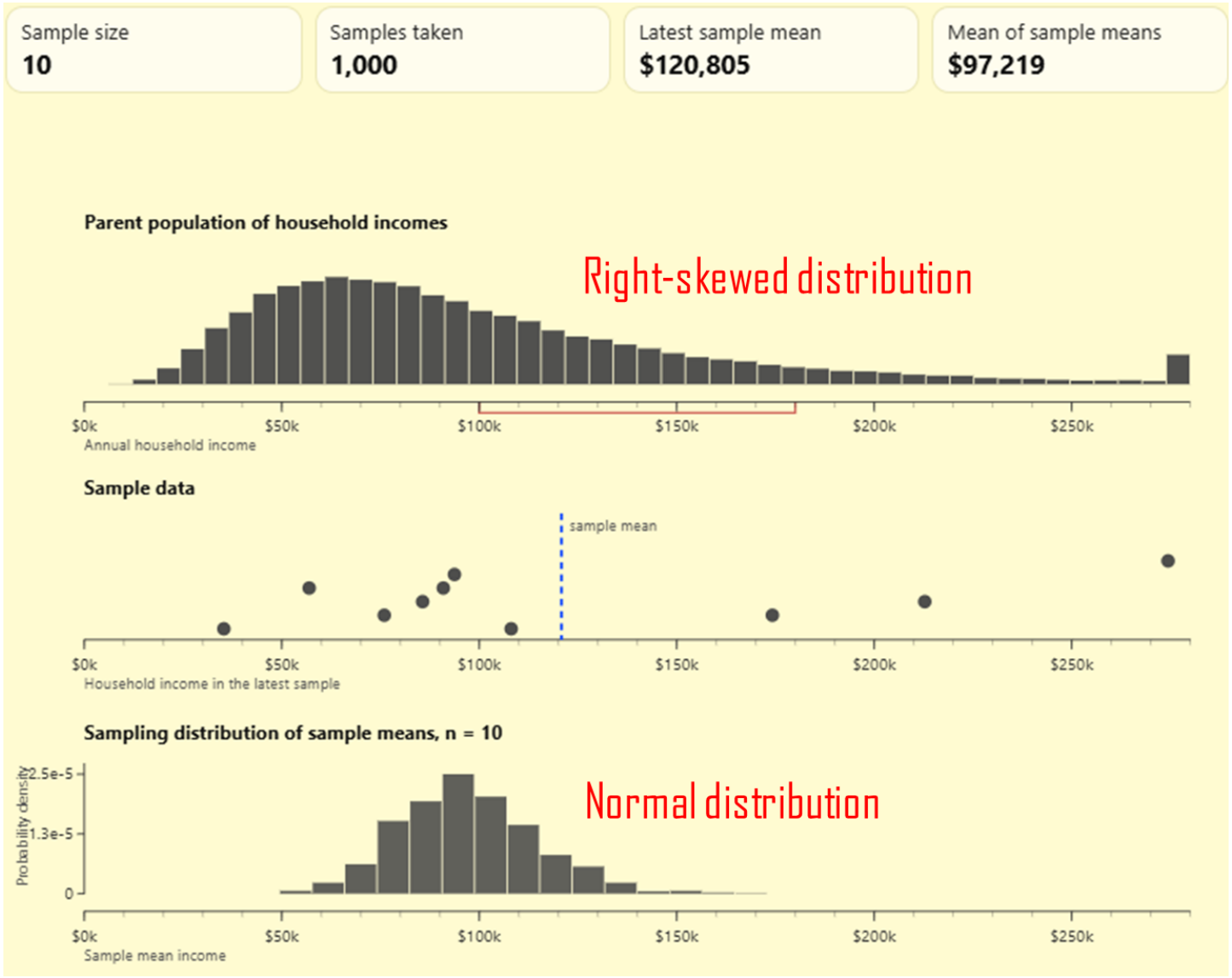
The sampling distribution tells us how much that estimate would vary if we could repeat the sampling process



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Central Limit Theorem

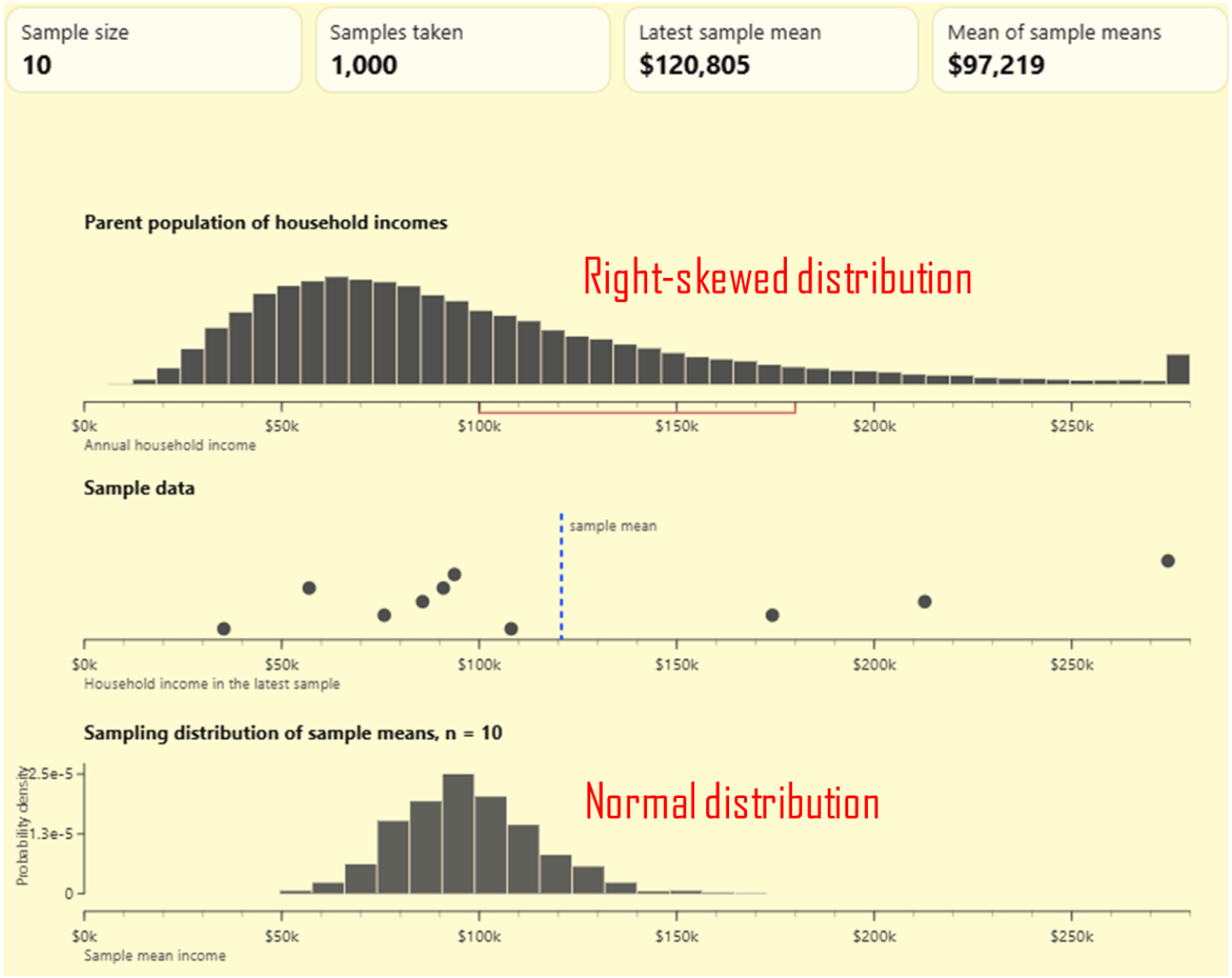
- The population of interest does not necessarily have to be normal for the sampling distribution of the mean to be approximately normal.



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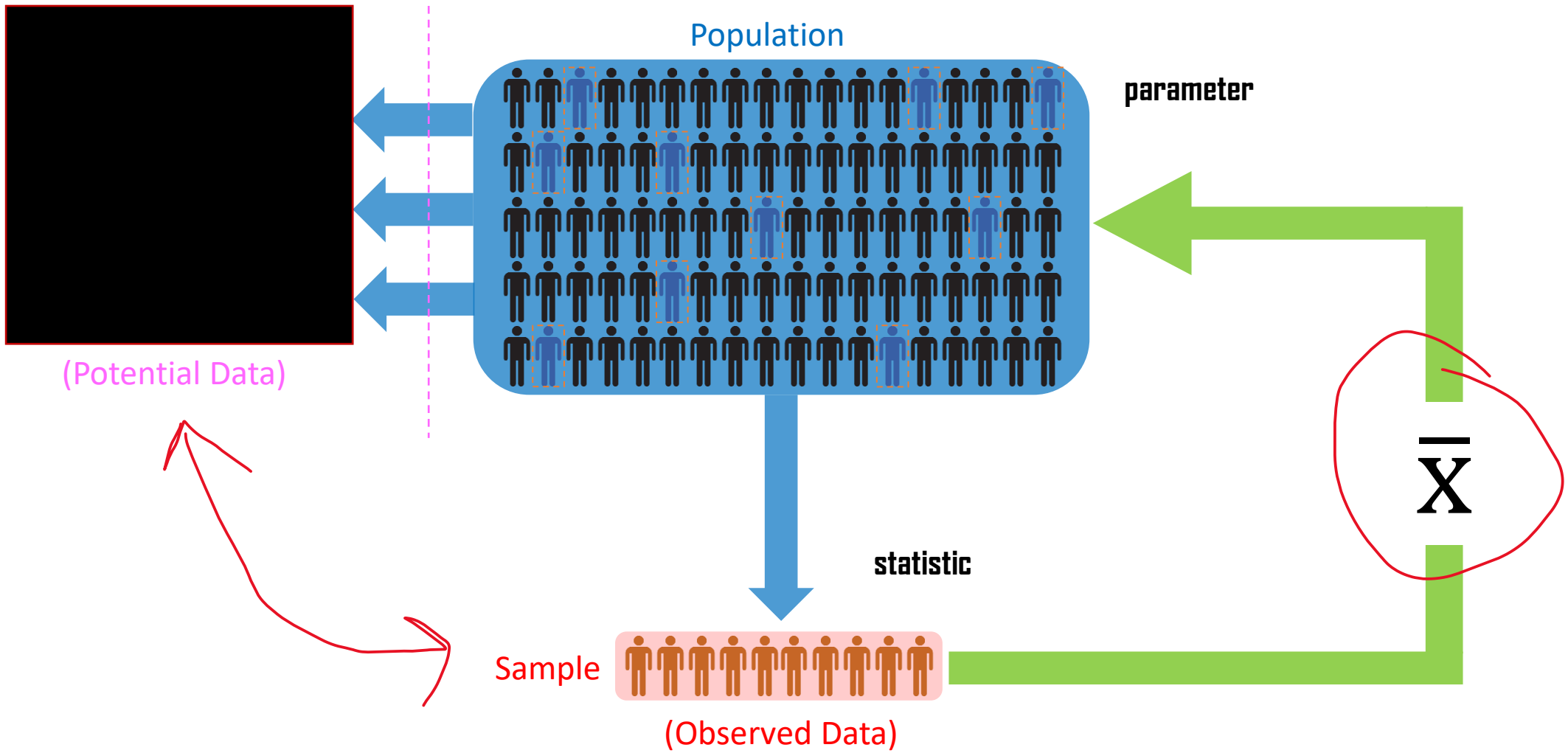
Central Limit Theorem

- The population of interest does not necessarily have to be normal for the sampling distribution of the mean to be approximately normal.



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Sampling Distribution



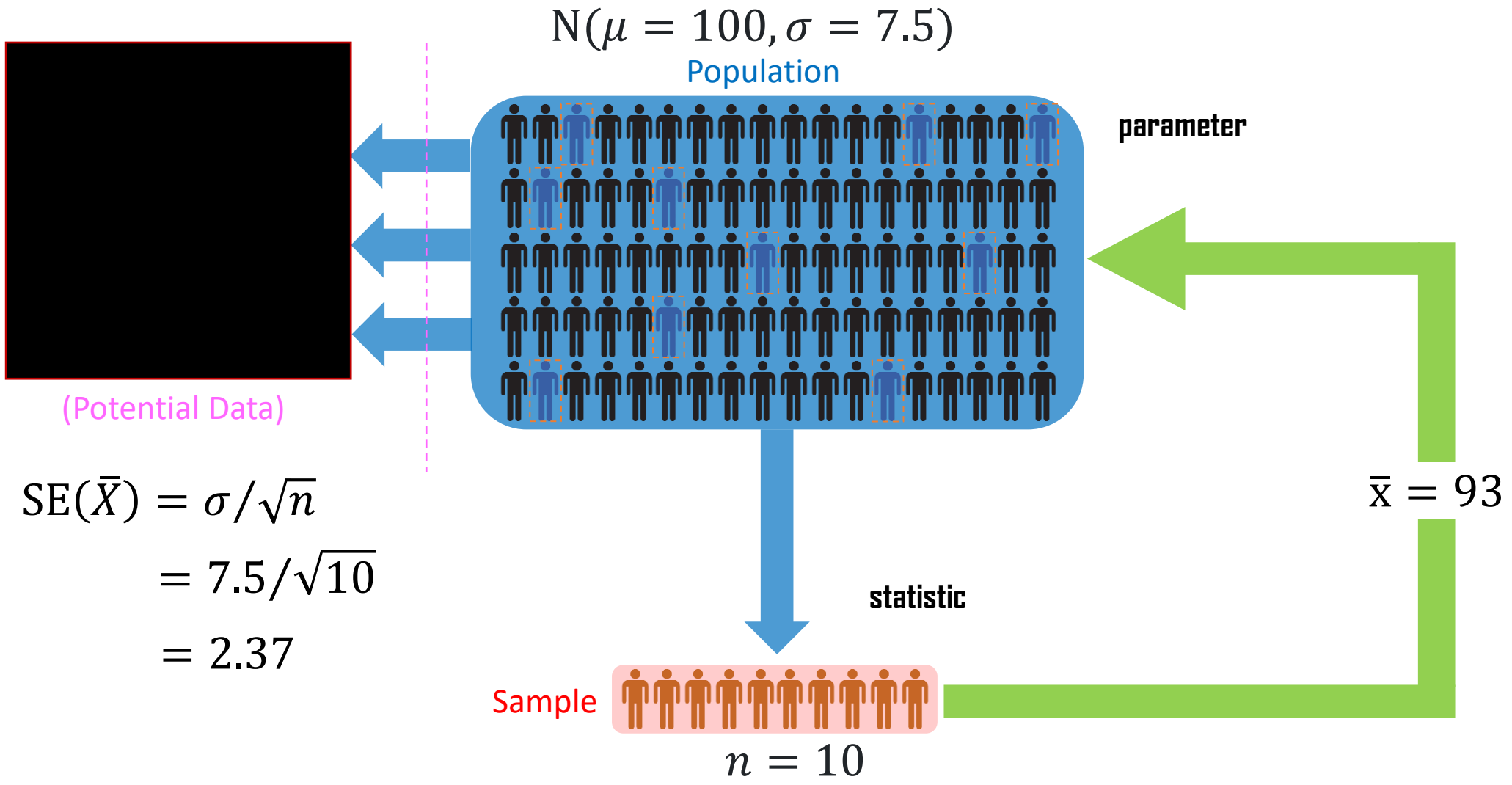
Standard Error

Shows how much the sample statistic itself varies across different potential samples

$$\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$$

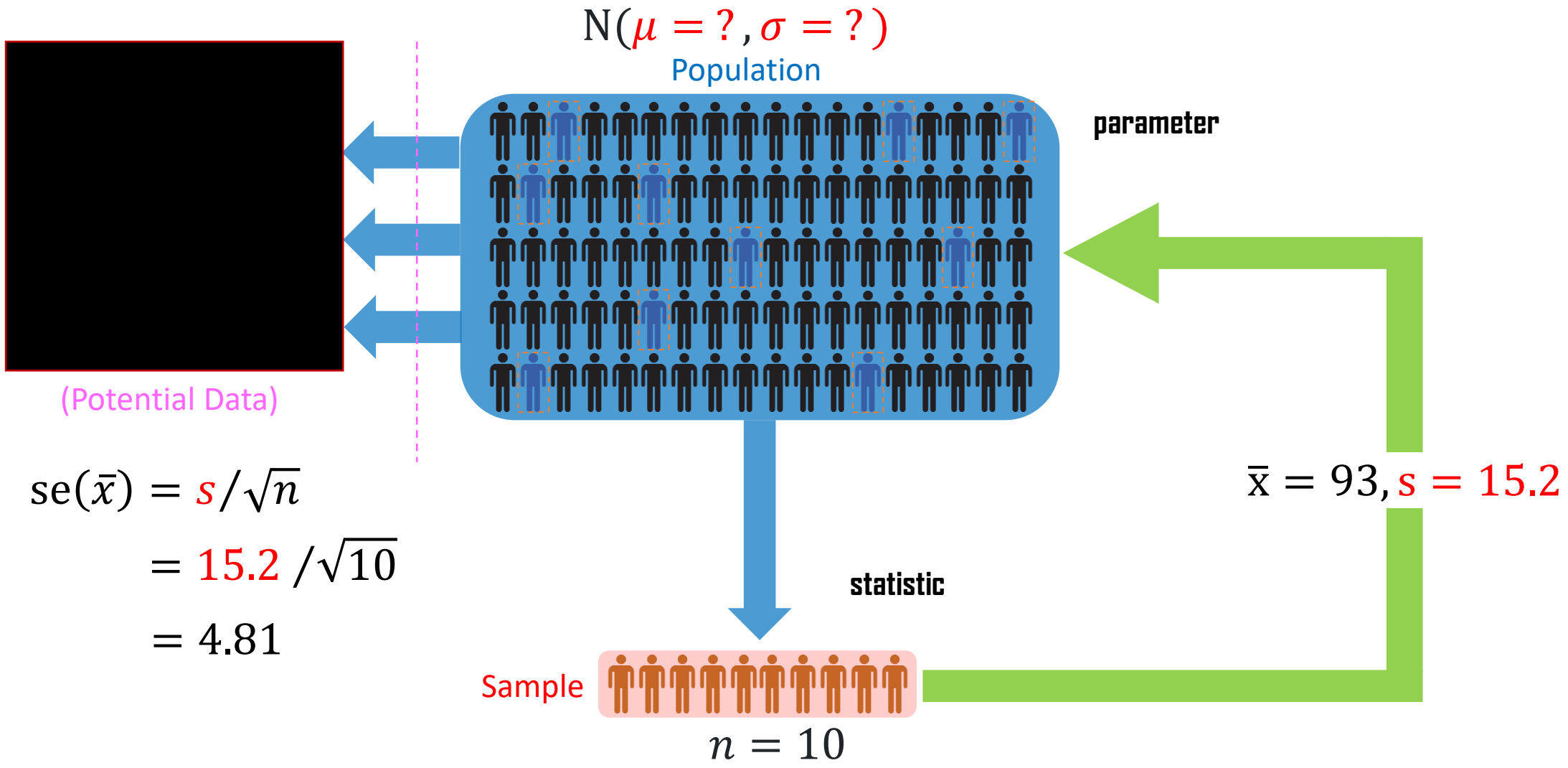
Where σ is the population std. deviation and n is the sample size

Standard Error – when σ is known



$$SE(\bar{X}) = \sigma/\sqrt{n}$$
$$= 7.5/\sqrt{10}$$
$$= 2.37$$

Standard Error – when σ is unknown



$$se(\bar{x}) = s/\sqrt{n}$$
$$= 15.2/\sqrt{10}$$
$$= 4.81$$

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Reporting estimates with SE

$$\text{estimate} \pm \text{std. error}$$

UNIVERSITY A

$$\hat{p} = .70 \pm 0.15$$



UNIVERSITY B

$$\hat{p} = .70 \pm 0.29$$

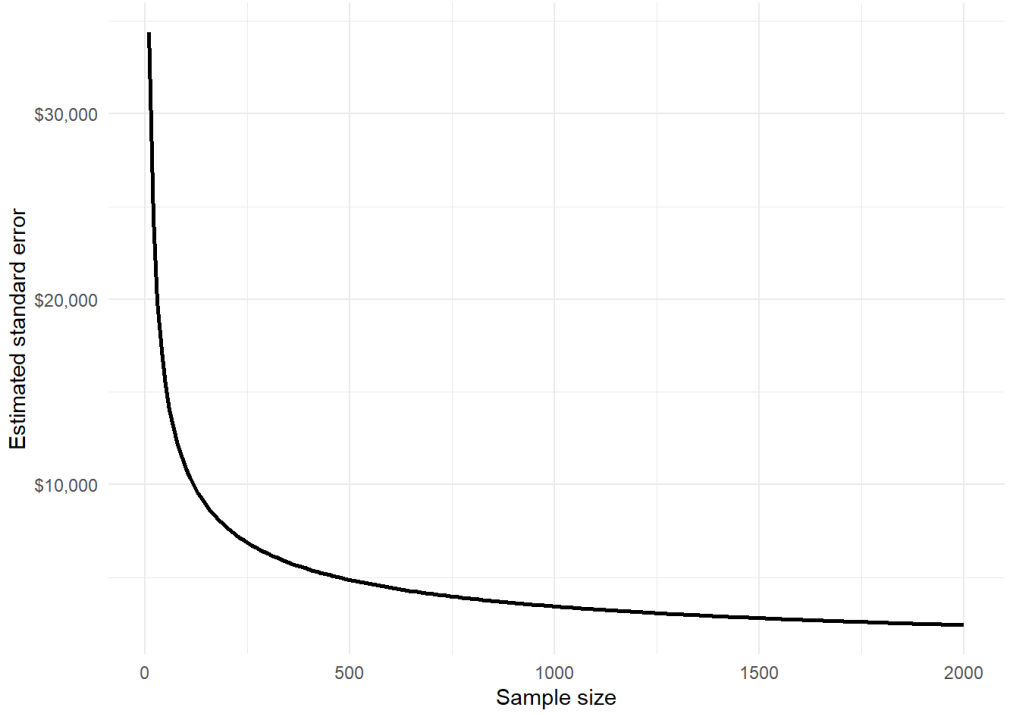


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Standard Error and Sample Size

Increasing the sample size reduces the standard error, but with diminishing returns.
To halve the standard error, we generally need about four times the sample size.

s	n	SE
10	10	$10/\sqrt{10} = 3.16$
10	20	$10/\sqrt{20} = 2.24$
10	40	$10/\sqrt{40} = 1.58$
10	60	$10/\sqrt{60} = 1.29$
10	100	$10/\sqrt{100} = 1$



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Point and Interval Estimate

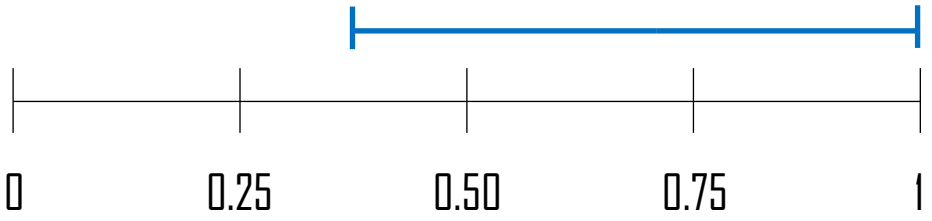
POINT ESTIMATE

A point estimate is a single numerical value calculated from sample data to serve as your best guess for an unknown population parameter (e.g. $\bar{x}, s, \hat{p}$, etc.)



INTERVAL ESTIMATE

An interval estimate is a range of values calculated from sample data that is likely to contain an unknown population parameter, e.g. confidence intervals, margin of error

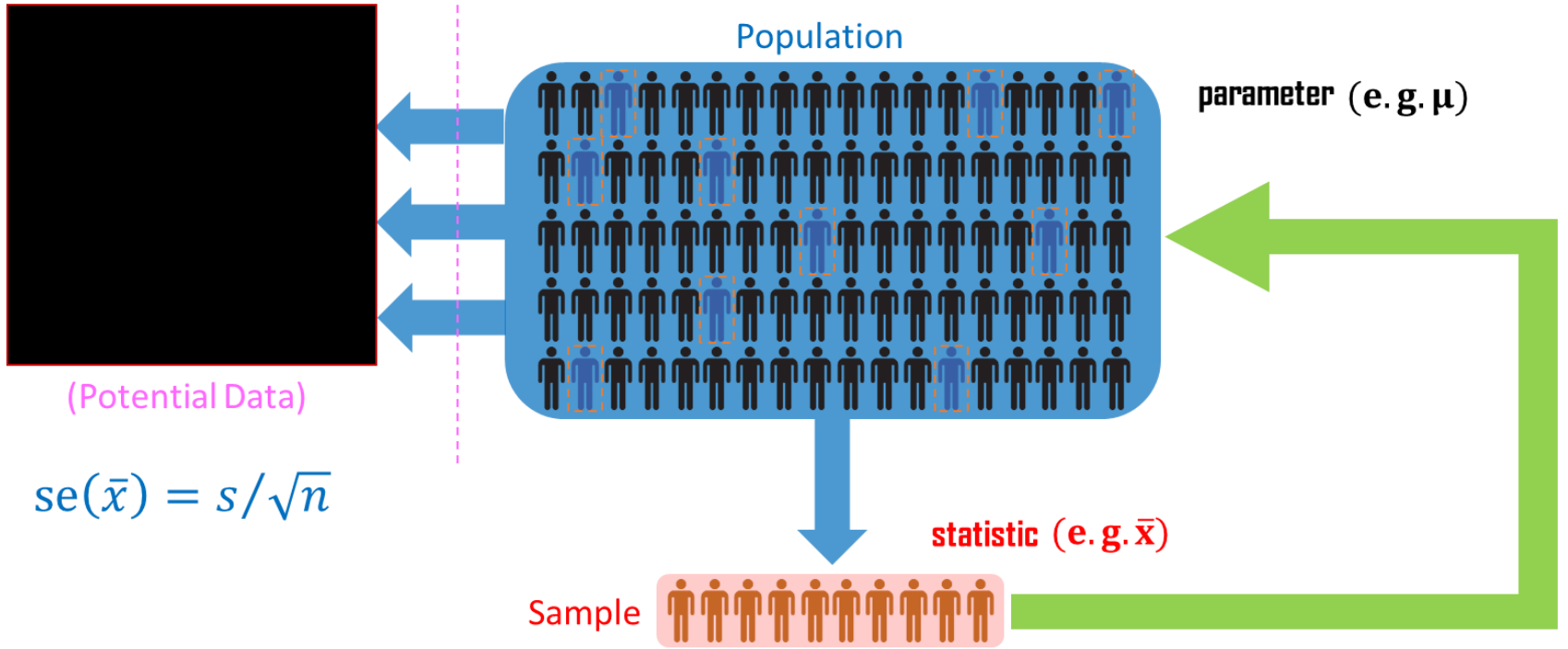


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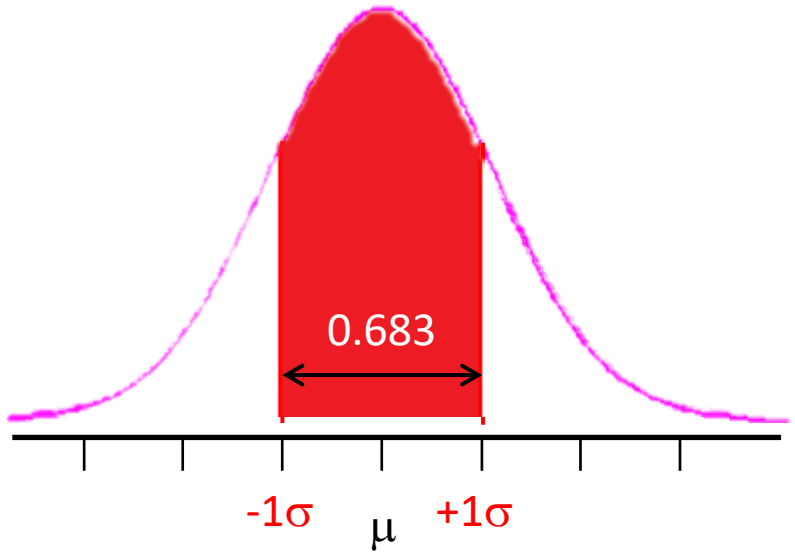
Confidence Interval

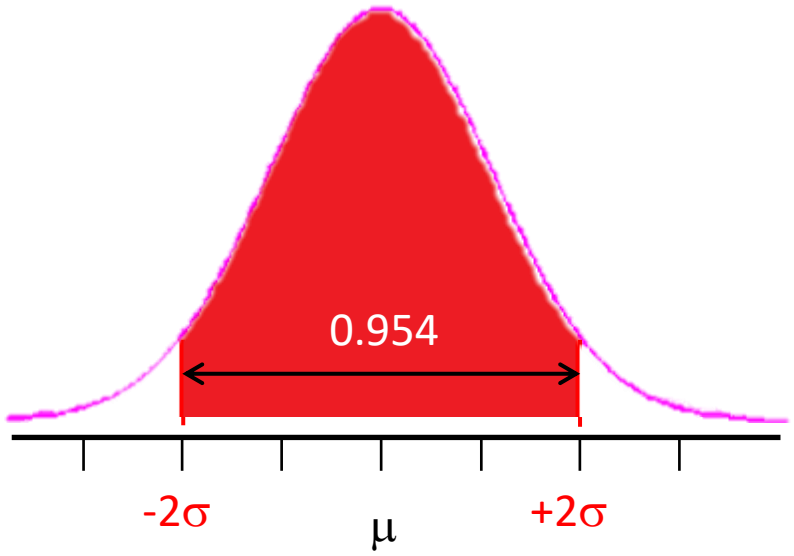
A confidence interval is a range of values which is likely to contain (in repeated sampling) the true value of an unknown statistical parameter, such as a population mean

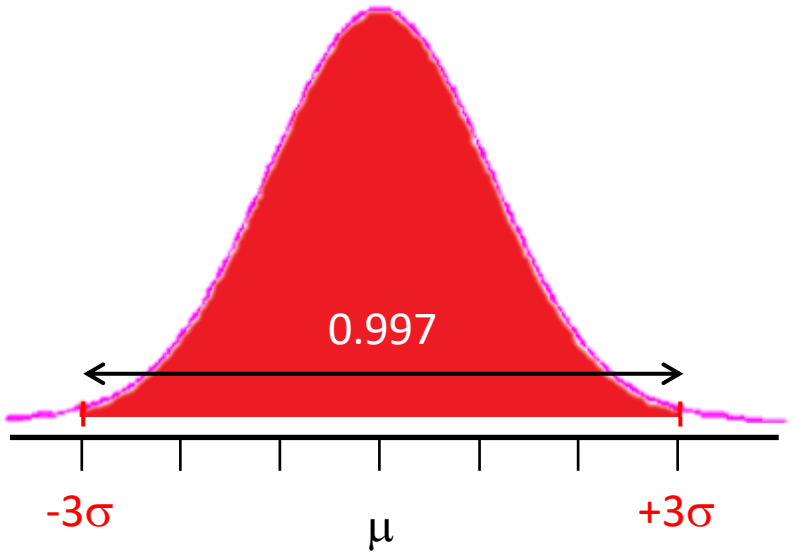
$$\text{estimate} \pm \text{multiplier} \times \text{std. error}$$



Confidence Interval

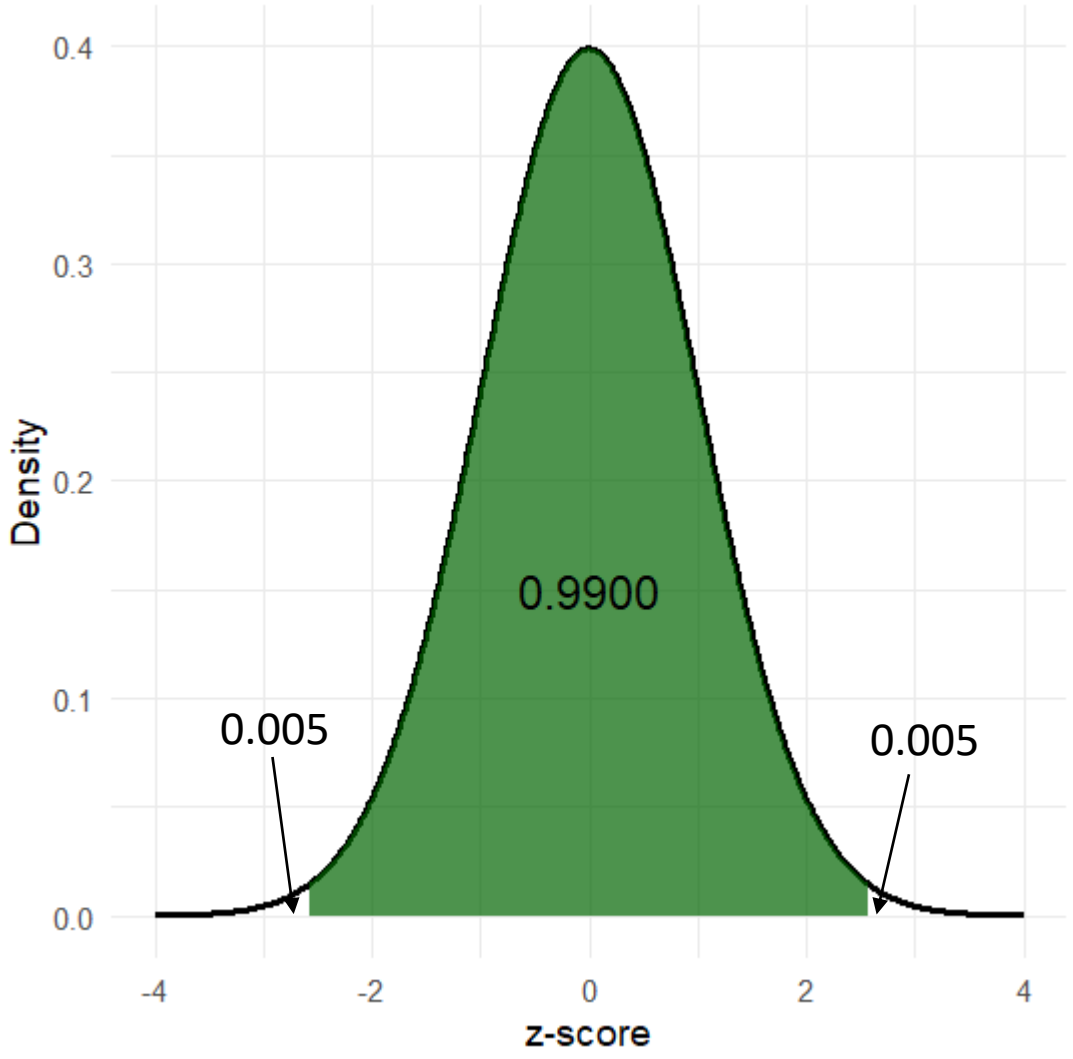






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Confidence Interval - multiplier

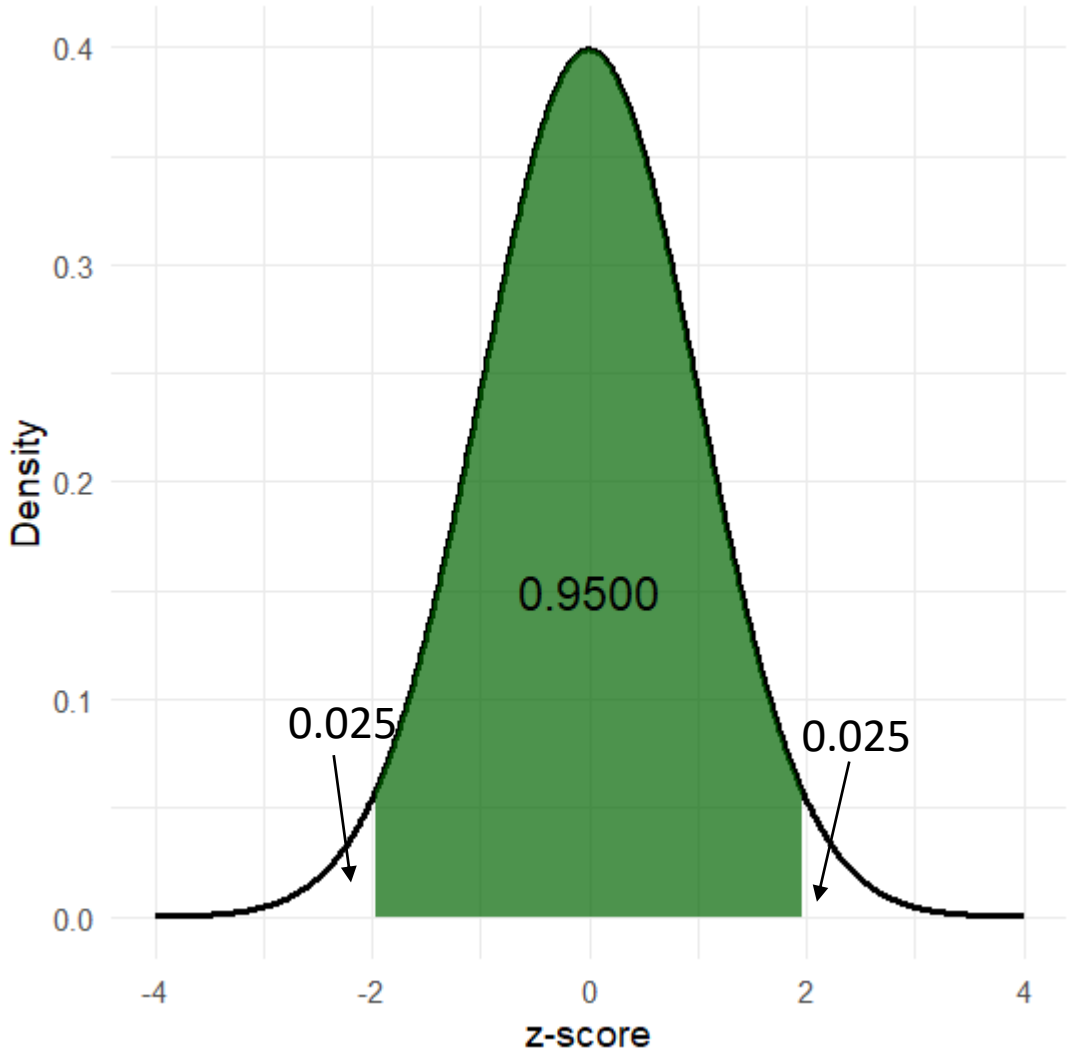


z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100

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$$99\%CI = \text{estimate} \pm 2.58 \times \text{std. error}$$

Confidence Interval - multiplier



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100

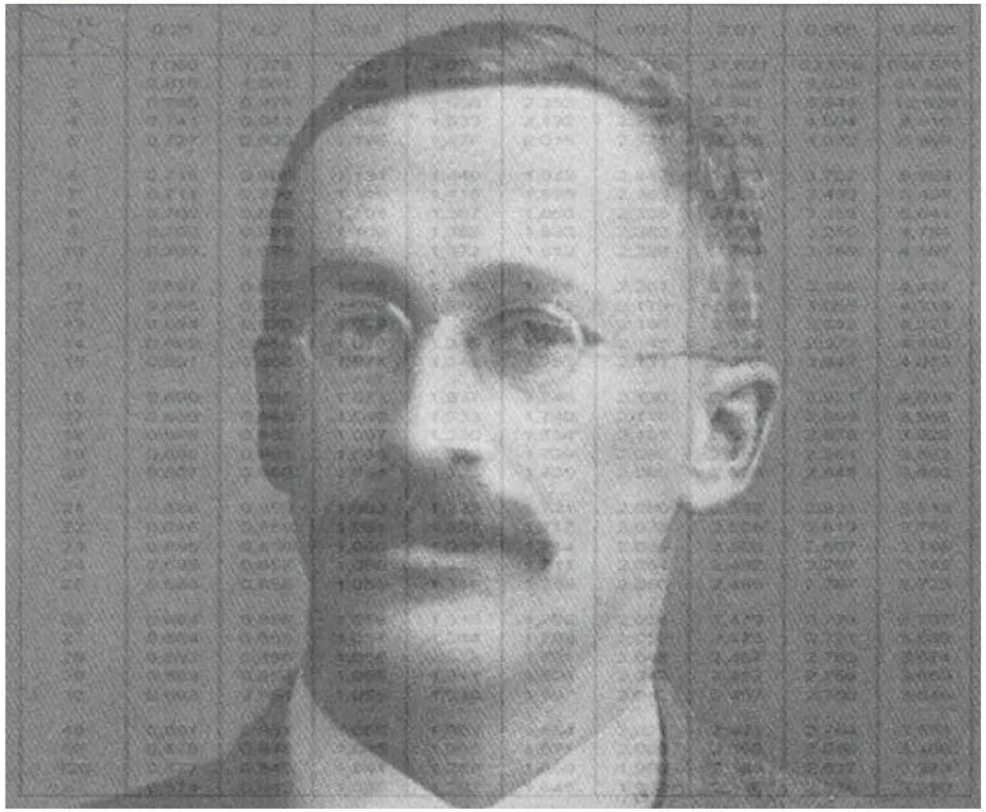
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$$95\%CI = \text{estimate} \pm 1.96 \times \text{std. error}$$

William Gosset – The t (student) distribution

The Guinness Brewer Who Revolutionized Statistics

Priceonomics



One of the greatest minds in 20th Century statistics was not a scholar. He brewed beer.

- Gosset was interested in small samples because Guinness often had to make decisions using relatively small amounts of data.
- In 1908, he published an article: "The Probable Error of a Mean"
- Because Guinness did not allow employees to publish under their own names, he published the paper under the pseudonym "Student." That's why we commonly call it Student's t-distribution.
- The t-distribution provides a way to account for the extra uncertainty from estimating the population standard deviation using the sample standard deviation.

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Degrees of Freedom (df)

- The df depend on the statistical procedure and how much information is available to estimate the parameters of the model. In general, degrees of freedom represent the amount of independent information available for estimating variability.

Statistical procedure	**Degrees of freedom**
One-sample t-test	$n - 1$
Paired-samples t-test	$n - 1$, where n is the number of pairs
Two-sample t-test (pooled)	$n_1 + n_2 - 2$
Welch's two-sample t-test	Approximate df based on n_1, n_2, s_1, and s_2
One-way ANOVA	Between groups: $k – 1$; Within groups: $N – k$
Chi-square test of independence	$(r-1) \times (c-1)$
Chi-square goodness-of-fit	$k - 1$
Simple linear regression	$n - 2$ for testing the slope
Multiple linear regression	$n - p - 1$, where p = number of predictors

Practice Questions – Confidence Intervals

Suppose we sampled 10 households from the Australian population to find out about the typical annual income. The results from this sample found $\bar{x} = 119{,}100$ and $s = 34{,}385$.

Calculate a 95% confidence interval of the population mean for this scenario. Let the multiplier = 2.26

Practice Questions – Confidence Intervals

In the previous question, we calculated a 95% confidence interval for the population mean as $94,525.91 to $143,674.10. Consider the following statement:

There is a 95% probability that the true population mean is between $94,525.91 and $143,674.10.

Is this statement true or false?